A proof of Hardin's empirical column recurrences for OEIS A245995, formalized in Lean 4

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Abstract

Let $T(n,k)$ be the number of arrays of length $n+2$ over $\{0,\dots,k\}$ in which no two entries among any three consecutive terms sum to exactly k (OEIS A245995 [1]). R. H. Hardin recorded nine column recurrences, for $k=1,\dots,9$, labelled "Empirical" in the OEIS. We prove a uniform statement for every k: for odd k, $T(n+2,k)=(k-1)T(n+1,k)+T(n,k)$; for even $k\geq 2$, $T(n+4,k)=(k-2)T(n+3,k)+T(n+2,k)+(k-1)^2T(n+1,k)+(k-1)T(n,k)$; and for $k=2$ the shorter $T(n+3,2)=T(n+2,2)+T(n,2)$. All nine of Hardin's formulas follow. The proof is a short transfer-matrix argument on four pair types and is machine-checked in Lean 4 with Mathlib [5]: no `sorry`, and only the axioms `propext`, `Classical.choice` and `Quot.sound`.

1 Introduction

OEIS A245995 (R. H. Hardin, 2014) tabulates $T(n,k)$, the number of words $w_1\cdots w_{n+2}$ over $\{0,\dots,k\}$ with $w_i+w_j\neq k$ whenever $1\leq j-i\leq 2$. The entry lists, as "Empirical for column k",

$$
\begin{aligned}
&k=1: a(n)=a(n-1), && k=2: a(n)=a(n-1)+a(n-3),\\
&k=3: a(n)=2a(n-1)+a(n-2), && k=4: a(n)=2a(n-1)+a(n-2)+9a(n-3)+3a(n-4),\\
&k=5: a(n)=4a(n-1)+a(n-2), && k=6: a(n)=4a(n-1)+a(n-2)+25a(n-3)+5a(n-4),\\
&k=7: a(n)=6a(n-1)+a(n-2), && k=8: a(n)=6a(n-1)+a(n-2)+49a(n-3)+7a(n-4),\\
&k=9: a(n)=8a(n-1)+a(n-2).
\end{aligned}
$$

We prove these together with the evident pattern for all k.

2 Related work

The table and its column recurrences were contributed by Hardin in 2014 [1]. Columns $k=2,\dots,7$ are the separate entries A245989–A245994 [2]. Colin Barker (2018) and R. J. Mathar (2014) added conjectured generating functions to several of these columns, still marked as conjectures or "Empirical" [2]. Column 2 (A245989) states its recurrence for $n\geq 5$ without the "Empirical" label. Row 1 (A245996, length-3 arrays) was proved by Pontus von Brömssen in 2022 by inclusion–exclusion [3]. To our knowledge no proof of the general column recurrences had been published; an OEIS and web search on October 12, 2026 found none. Counting words that avoid a finite set of local patterns by a transfer matrix is classical (see e.g. Stanley [4], §4.7), so the method is standard; what is new is the four-state reduction that works uniformly in k, and the machine-checked proof. The formalization uses Lean 4 [6] and Mathlib [5].

3 Statement

Theorem 1. *For all $n \geq 0$:*

1. *if k is odd, $T(n+2,k) = (k-1)T(n+1,k) + T(n,k)$;*
2. *if $k \geq 2$ is even, $T(n+4,k) = (k-2)T(n+3,k)+T(n+2,k)+(k-1)^2T(n+1,k)+(k-1)T(n,k)$;*
3. *$T(n+3,2) = T(n+2,2) + T(n,2)$ and $T(n+1,1) = T(n,1)$.*

Here $T(0,k)$ counts words of length 2.

Corollary 2. *All nine of Hardin's column recurrences hold for every n at which they are defined (OEIS offset $n \geq 1$).*

4 Proof

Write $\bar{a} = k - a$, so a pair is *forbidden* iff $y = \bar{x}$. Let $\varepsilon = 1$ if k is even and 0 otherwise; when k is even, $h = k/2$ is the unique letter with $\bar{h} = h$. Let $N_L(x,y)$ be the number of good words of length $L+2$ beginning x, y.

Step 1 (prepending a letter). A word $c\,v$ is good iff v is good and $c + v_1 \neq k$, $c + v_2 \neq k$. Hence

$$N_{L+1}(c,b) = [\,b \neq \bar{c}\,] \sum_{a \neq \bar{c}} N_L(b,a).$$

Step 2 (four types). Call a non-forbidden pair (x,y) of type E if $x = y$, P if $x = h$, Q if $y = h$, and D otherwise. We claim $N_L(x,y)$ equals E_L, P_L, Q_L, D_L according to the type (and 0 if forbidden), where $E_0 = D_0 = P_0 = Q_0 = 1$ and, with

$$R_L = E_L + \varepsilon Q_L + (k - 1 - \varepsilon) D_L,$$

$$E_{L+1} = R_L, \quad D_{L+1} = R_L - D_L, \quad P_{L+1} = R_L - Q_L, \quad Q_{L+1} = (k-1)P_L.$$

Indeed, for $b \neq h$ the row sum $\sum_a N_L(b,a)$ is R_L: the column $a = \bar{b}$ contributes 0, $a = b$ contributes E_L, $a = h$ (if it exists) contributes Q_L, and the remaining $k - 1 - \varepsilon$ letters contribute D_L each. For $b = h$ the row sum is kP_L. By Step 1, $N_{L+1}(c,b)$ is the row sum of b minus the single term $N_L(b,\bar{c})$, and a check of the four cases gives the recursion above. For example, if $c = h \neq b$ then $N_L(b,\bar{c}) = N_L(b,h) = Q_L$; if $b = h \neq c$ then $kP_L - P_L = (k-1)P_L$.

Step 3 (total). Summing rows, $T(L,k) = \varepsilon k P_L + (k + 1 - \varepsilon) R_L$.

Step 4 (recurrences). The vector (E, D, P, Q) evolves by a fixed linear map M. Theorem 1 is the statement that the linear functional T annihilates $M^2 - (k-1)M - I$ (odd k, where $\varepsilon = 0$ and only E, D enter), $M^4 - (k-2)M^3 - M^2 - (k-1)^2 M - (k-1)I$ (even k), and $M^3 - M^2 - I$ (at $k = 2$). Each is a polynomial identity in k, E, D, P, Q, verified by direct expansion (in Lean, by `ring`). Corollary 2 is the specialisation $k = 1, \ldots, 9$, shifted to the OEIS indexing.

Numerical check. The recursion reproduces all 9999 terms of Hardin's b-file.

5 Formalization

All declarations are in namespace `A245995`; see Table 1.

Lean name (file)	Statement
`Good`, `T`, `N` (`Basic.lean`)	OEIS condition (as natural-number sums); $T(n,k)$; $N_L(x,y)$
`good_cons`, `N_succ`	Step 1
`typ`, `V`, `row_sum`, `N_eq_typ` (`Types.lean`)	Step 2
`T_eq_sum`, `T_eq_F` (`Main.lean`)	Step 3
`T_odd`, `T_even`, `T_two`, `T_one`	Theorem 1
`T_col1`, ..., `T_col9`	Corollary 2, verbatim OEIS form
`Values.lean`	kernel-checked values $T(1,2)=8$, $T(2,3)=68$, $T(0,4)=20$

Table 1: Theorems and their Lean names.

6 Reproducibility

- Source code: `https://github.com/het4rk/OEIS-A245995-proof`. The Lean project is in `A245995/`.
- Toolchain: `leanprover/lean4:v4.35.0-rc4`. Mathlib: tag `v4.35.0-rc4`, commit `1f414401f69059aa7eead47b53ee40bd38455eeb`, pinned in `lake-manifest.json`.
- Build:

```
cd A245995
lake exe cache get
lake build
lake env lean Axioms.lean
```

- Axioms: every theorem listed in `Axioms.lean` (`T_odd`, `T_even`, `T_two`, `T_one`, `T_col1`–`T_col9`) depends only on `propext`, `Classical.choice` and `Quot.sound`.
- Attestation: `verification/` contains the build log, the axiom output and `VERIFICATION.md` with toolchain details and SHA-256 hashes of the Lean sources. Continuous integration rebuilds on every push, rejects `sorry` and non-standard axioms, and uploads the logs as artifacts.

7 Limitations

The argument is a routine transfer-matrix computation; its interest is that it settles a family of empirical OEIS formulas uniformly in k, with a machine-checked proof. We do not formalize Hardin's empirical *row* recurrences (fixed n, varying k), apart from row 1, which was already proved [3] and is not restated here. We do not prove that the recurrences are of minimal order for every k, nor the conjectured generating functions, although these follow from the recurrences and initial terms.

References

[1] R. H. Hardin, *A245995: $T(n,k)$ = number of length $n+2$ $0..k$ arrays with no pair in any consecutive three terms totalling exactly k*, The On-Line Encyclopedia of Integer Sequences, 2014, `https://oeis.org/A245995`.

[2] R. H. Hardin, C. Barker, R. J. Mathar, *A245989, A245990, A245991, A245992, A245993, A245994* (columns $k = 2, \ldots, 7$ of A245995), The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A245990`.

[3] P. von Brömssen, proof in *A245996* (row 1 of A245995), The On-Line Encyclopedia of Integer Sequences, 2022, `https://oeis.org/A245996`.

[4] R. P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge University Press, 2012.

[5] The mathlib Community, *The Lean mathematical library*, in Proc. CPP 2020, ACM, 367–381.

[6] L. de Moura, S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, LNCS 12699, Springer, 2021, 625–635.